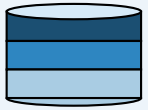
CARDIO TOXICITY ML PIPELINE & DEPLOYMENT ARCHITECTURE

End-to-End Workflow: Data Ingestion -> Preprocessing -> Stacking Ensemble -> Conformal Calibration -> Gradio UI

STAGE 1: Data & Preprocessing

28 CiPA Benchmark Drugs



2,000 cells/drug

12 Train Drugs (24,000 cells)

16 Test Drugs (32,000 cells)

Data Audit & Cleaning

Sentinel check ('#VALUE!')

Dropped 26 corrupted rows
(1.25% mexiletine, 0.05% ibutilide)

Audit Log: DROPPED_ROWS_REV.csv

11 Canonical Biomarkers

qNet	max_dv
dvdtmax	camax
vmax	carest
vrest	CaTD50
APD50	CaTD90
APD90	

**Pre-normalized by provider (no StandardScaler)*

STAGE 2: Grouped CV

StratifiedGroupKFold
(k = 4, by drug_name)

Fold 1: 9 Train / 3 Val

V T T T

Fold 2: 9 Train / 3 Val

T V T T

Fold 3: 9 Train / 3 Val

T T V T

Fold 4: 9 Train / 3 Val

T T T V

Guarantees zero drug leakage for out-of-fold meta-features

STAGE 3: Stacking Ensemble

Level 1: Base Learners

Random Forest Classifier

n_estimators=400, criterion='entropy'

Outputs: [p_high, p_int, p_low]_RF

XGBoost Classifier

n_estimators=300, lr=0.1, max_depth=10

Outputs: [p_high, p_int, p_low]_XGB

Level 2: Meta-Learner (Blender)

Logistic Regression Meta-Learner

C=1.0, max_iter=1000

Produces smooth, calibrated probabilities!

**Fixes 100%/0% step-function decision tree artifacts*

STAGE 4: Calibration & App

Cross-Conformal Calibration

Leave-One-Drug-Out (LODO)
over 12 training drugs

Threshold q_hat = 0.6275

(11th of 12 LODO scores)

Nominal Coverage: 84.6%

Hugging Face Space App

mitcheltastic/tdp-cipa-screening

Conformal Prediction Set

C(X) = {c: p_c >= 1 - q_hat}

Clinical Safety Action

Most severe class in C(X)

SHAP Feature Attribution